

# Capacity is an interval: the traffic rule sets the floor, the network sets the ceiling

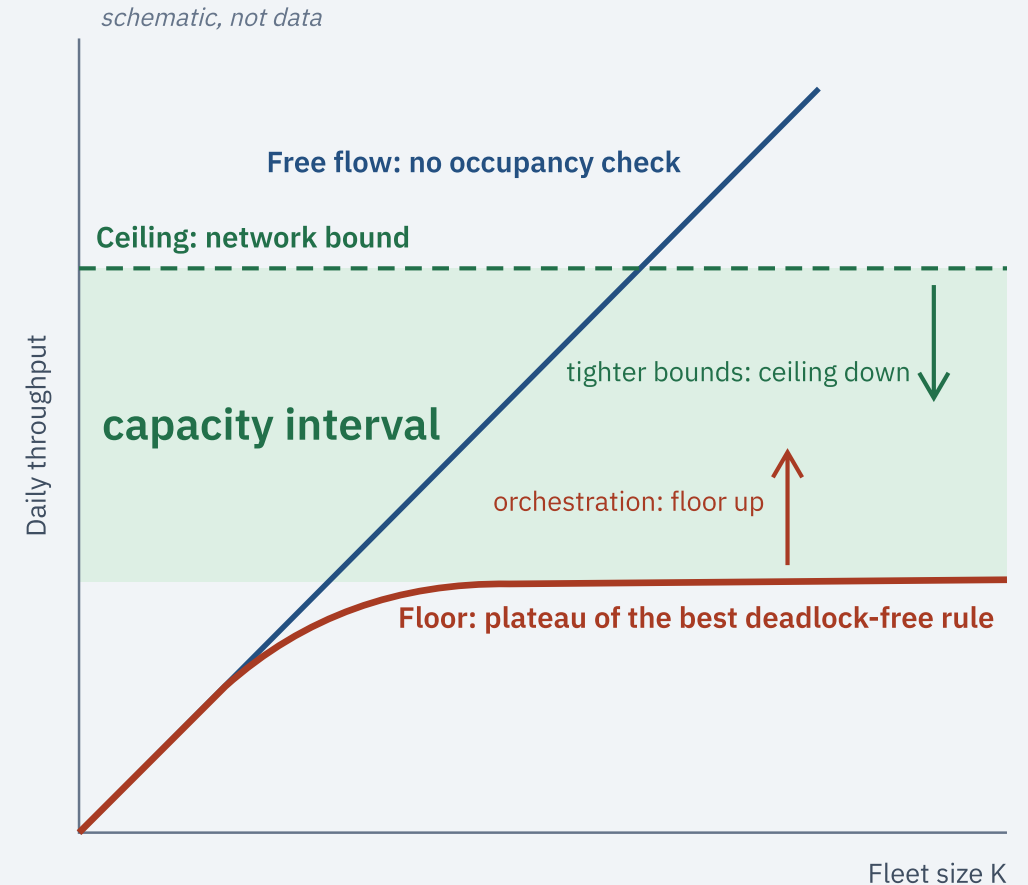
The flag experiment on two real yards: the capacity interval, the fleet–throughput trapezoid, the network-bound regime and the free-flow error

*When transporters own the road: capacity bounds and operating regimes in shipyard block logistics (working title)*

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Data: simulations in this study (flag experiment to system bound, all verified); literature anchors checked in full text



# The sub-study in one minute

1

## Question

When loaded vehicles own the road, how much can block transport deliver, when do more vehicles stop helping, and how wrong is free flow?

2

## Method

Two real networks, Okpo and Yantai; four traffic rules, two granularities; fleets up to 150, 10 seeds each; an LP gives the network bound; vehicles needed at whole-yard volume, 16-h and 24-h days.

3

## Results so far

Capacity intervals: Okpo [627, 1,720], Yantai [455, 1,252], the floor from the best deadlock-free rule at 36% of the ceiling on both. At Okpo's own 339 a day reservation needs 27 vehicles (16 h), below  $K^* 50-60$ ; at the peak of 678, 117.

4

## Next

A less conservative deadlock-free rule is under test (floor, decline hypothesis); next, the network-bound hypothesis's main caliber (16 h or 24 h, mean or peak); then retrofits, exchange rate, floating docks, holding areas.

**Capacity is not one number but an interval: the traffic rule sets its floor, the network its ceiling.**

# This sub-study answers two claims and two hypotheses: two ceilings, the trapezoid law, network-bound and free-flow error

|                            | Wording in this study (9 Oct 2026)                                                                                                                                                                                                                                                 | What this sub-study delivers                                                                                                                                                                   |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Two ceilings               | Throughput $\leq \min\{\text{fleet bound } K \cdot H / \bar{c}, \text{network bound}\}$ . The network bound is set by the blocking-time structure and can be computed by linear programming. Attainable capacity lies within the interval [best deadlock-free rule, network bound] | Intervals for both yards, both granularities and the erection mix, the floor from the best tested deadlock-free rule; the system and route-free bounds firm up the ceiling                     |
| Trapezoid law              | Throughput first rises linearly with fleet size, then enters a plateau set by the network; when waiting occupies the road, it falls beyond a jamming threshold. The free-flow model sees neither of the last two parts                                                             | First two parts: both yards' flag curves; the fall belongs to the decline hypothesis: teleport-free, deadlock-free safe release rises then falls, now re-checked with a less conservative rule |
| Network-bound hypothesis   | At a reasonable daily task volume and a realistic fleet size, the system is in the network-bound regime, at least at erection peaks or near the dock mouth                                                                                                                         | $K^*$ ; integer vehicles needed (16 h and 24 h) judged against the $K^*$ range; the density criterion; the work-zone chart; dock-mouth thresholds by fleet size                                |
| Free-flow error hypothesis | The free-flow model's underestimate of completion time and of vehicles needed grows superlinearly with vehicle density                                                                                                                                                             | The exponent $\alpha$ of the service-time gap $\Delta(K)$ ; the ratio of vehicles needed at whole-yard volume (integer $K$ , 16 h and 24 h)                                                    |

Source: the claims and hypotheses of this study (version of 9 Oct 2026), quoted verbatim in Chinese; English and Korean are faithful translations. Place in the thesis: chapter 3, capacity.

# The flag experiment: two real yard layouts, four traffic rules, two resource granularities, one task stream

|                         |                                                                                                                                                                    |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Free flow               | No occupancy check; vehicles never interact (the field's default)                                                                                                  |
| Whole-route reservation | Books the whole route before each loaded trip and waits off the road; deadlock-free by construction. Best tested such rule: the floor                              |
| Safe release            | Grants segment by segment after a banker's-algorithm safety check; no teleport, no deadlock; exact but conservative (new this round)                               |
| Segment request         | One segment at a time; on deadlock a vehicle leaves the network for 900 s<br><b>Reference rule (teleport clearing): supports no claim or hypothesis on its own</b> |

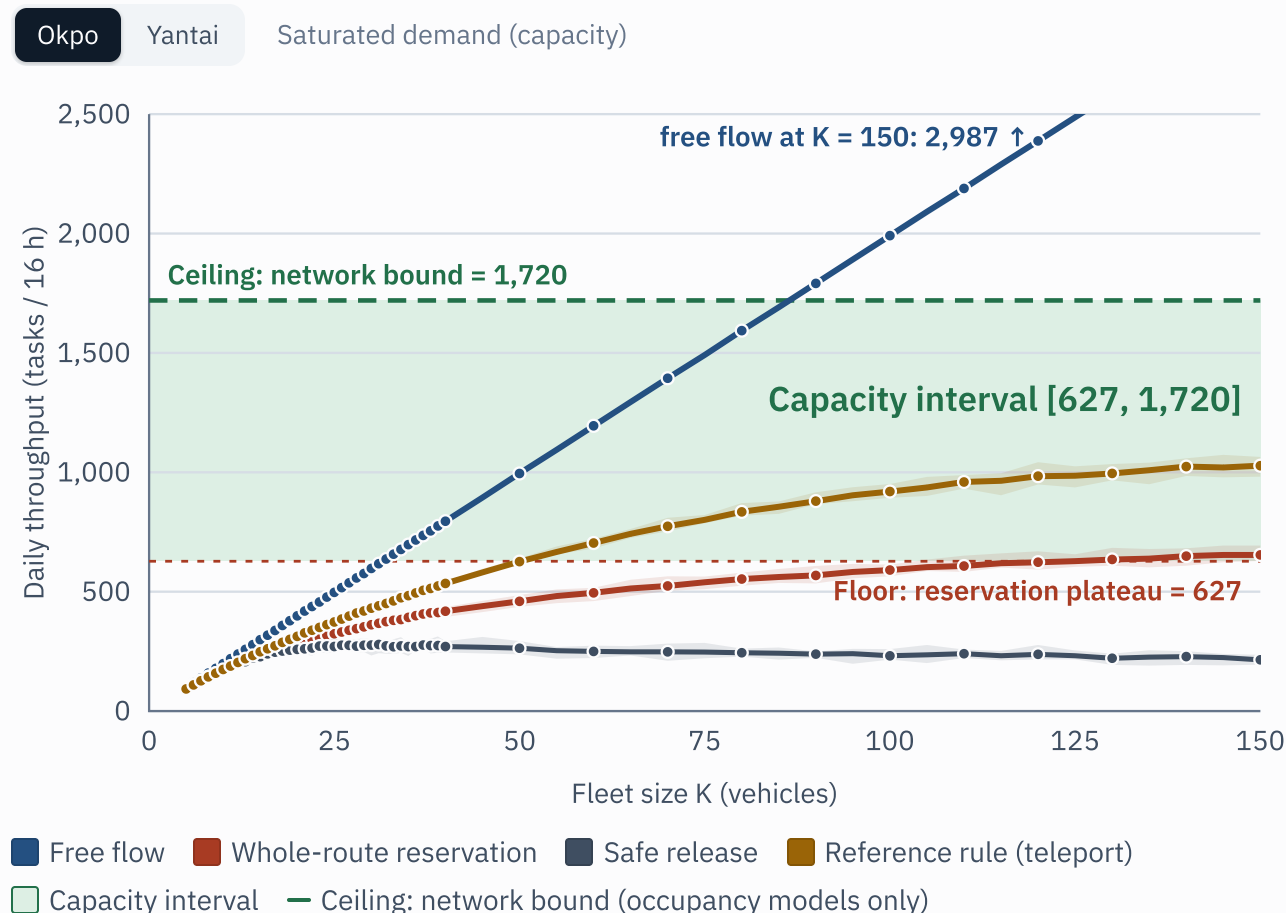
- **Demand:** saturated to measure capacity (K = 5–150, Yantai 1–150, 10 seeds, 4 days); finite whole-yard volumes of 300–1,200 a day plus Okpo's own 339 and peak 678, on 16-h and 24-h days; 97 / 194 is a four-transporter subset, for comparison only

|                                       | OKPO                                      | YANTAI                                          |
|---------------------------------------|-------------------------------------------|-------------------------------------------------|
| Roads (count / km)                    | 721 / 40.3                                | 140 / 8.6                                       |
| Junction resources                    | 334                                       | 58                                              |
| Stops                                 | 421                                       | 140                                             |
| Own daily volume: mean / peak         | 339 / 678                                 | 23 / 46                                         |
| Where the network bound binds (seeds) | dock-1 road 5/10, a through corridor 4/10 | the two doors of the paint shop (presumed) 9/10 |

- **Ceiling, the network bound:** an occupancy LP with width bin-packing rows (Fekete–Schepers), exact rationals; checked case by case, none exceeds it (17,760 runs earlier, 40,157 more this round)
- Maps and granularity:** Hanwha Ocean Okpo derived map before floating docks and branch roads and CIMC Raffles Yantai map (networks as in earlier rounds), traced on public imagery; neither is shown. Whole road: each road one resource; segmented: dock roads blocked at their stops (one rule for both yards); the imagery shows segmented is closer to Okpo's practice, so both are reported. Next round: the common yard-map dataset (Okpo with floating docks and branch roads).

Source: simulations in this study (settings of the flag-experiment, tightened-bound, crane-cadence, gantry-span and physical-rule rounds); speeds, platform size and handling times as set for the flag experiment; the map switch applies from the next round. Conditions: Okpo map before the floating docks and branch roads, Yantai map.

# The capacity interval [best deadlock-free rule, network bound]: Okpo [627, 1,720], Yantai [455, 1,252]; on both the floor is only 36% of the ceiling



## [627, 1,720]

Capacity interval (a day): the floor is the reservation plateau over K = 100–150, the ceiling the network bound (mean of 10 seeds, 1,436–1,948 across seeds)

## 36%

The floor reaches only this share of the ceiling: the gap is the traffic rule, which orchestration aims to close

## 229<sub>279 @ 31</sub>

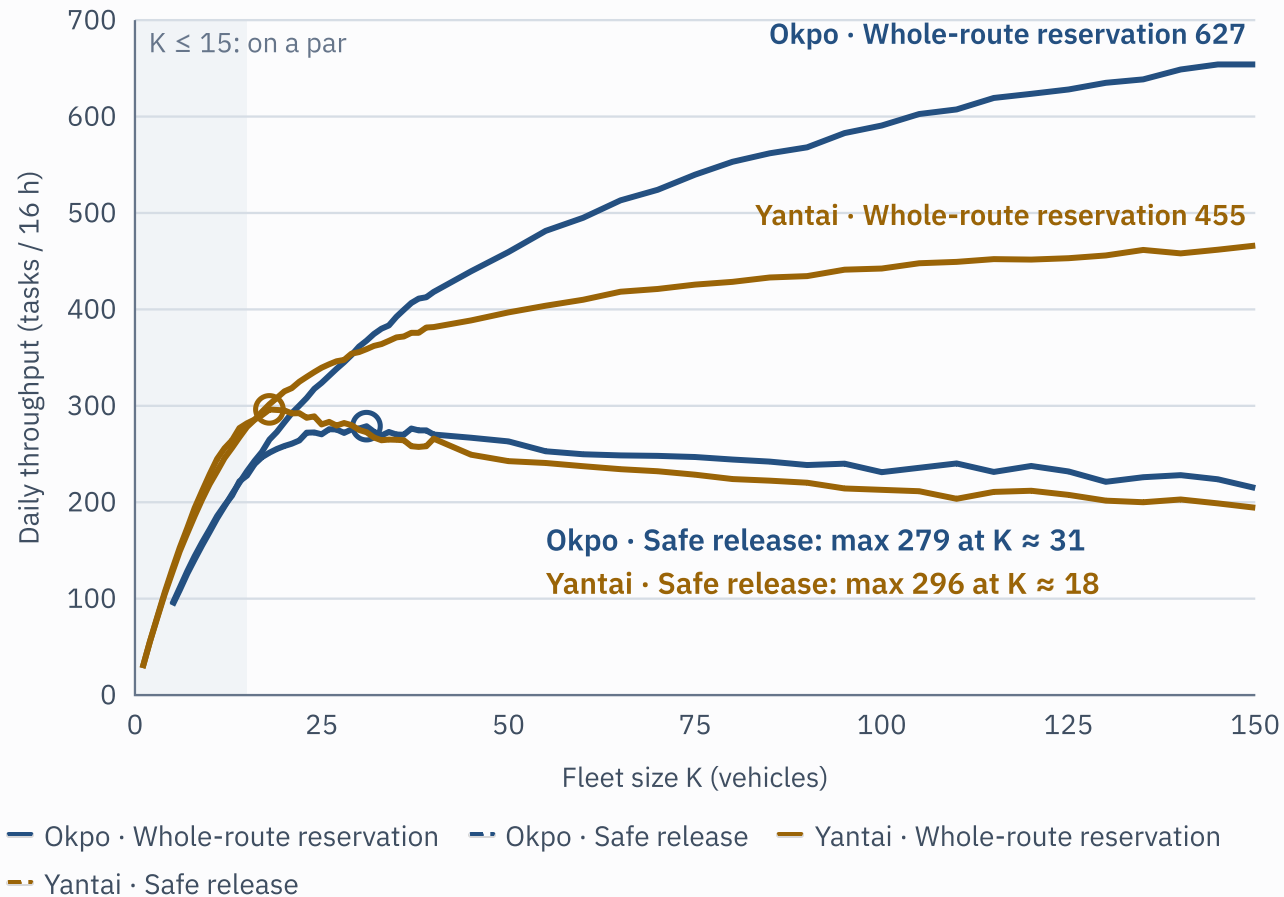
Safe-release plateau (no teleport, never deadlocks); maximum 279 at K ≈ 31, then falling, below reservation (next slide)

## 984<sub>57%</sub>

Reference-rule plateau (teleport clearing), for reference only

Source: simulations in this study on Okpo and Yantai, with batches for free flow, whole-route reservation, the reference rule and safe release. Plateaus are means over K = 100–150; curves are means of 10 seeds, shading min–max. Conditions: saturated demand, each road one resource; Okpo map before the floating docks and branch roads, Yantai map. The network bound constrains only the occupancy models; free flow checks no occupancy and may cross it.

# The floor is reservation: safe release is exact and never deadlocks, but conservative, and lower from $K = 20$



## Exact, never deadlocks

Its decisions match a naive reduction on the same input in all 860,291 checks; not one deadlock in any batch, and no vehicle leaves the network.

## Conservative claims

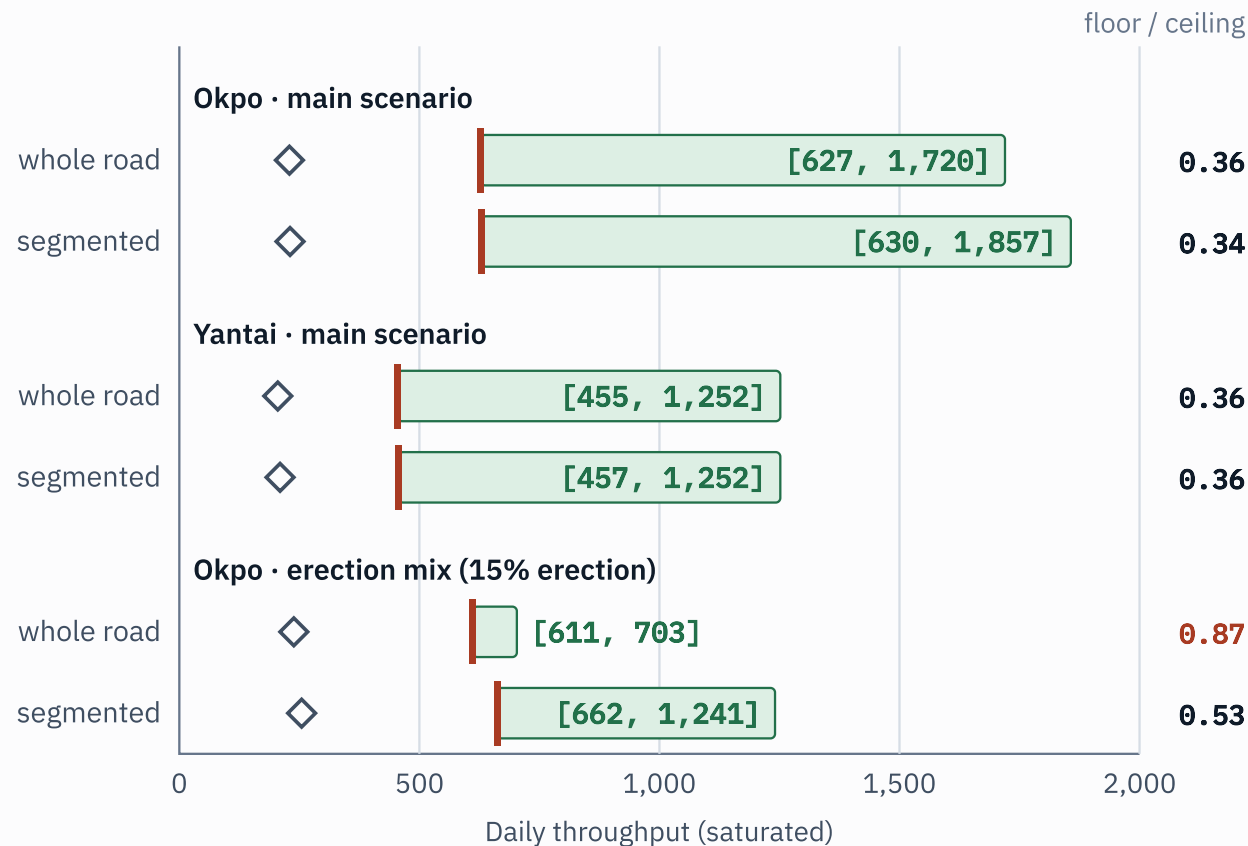
It claims each vehicle's whole remaining task and ignores timing: about 2/3 of its waits start from a rejection by the criterion; in the six-entrance ring case three vehicles following each other round a ring, never meeting, are still refused.

## Result and next step

On a par with reservation at  $K \leq 15$ , below it from  $K = 20$ , so both floors use reservation. Next, less conservative rules are tested: the banker's check plus a step-wise feasibility search, or claims cut at the next safe holding point (the safe-orchestration theorem).

Source: simulations in this study: saturated batches of safe release and whole-route reservation; decisions compared one by one with a naive reduction in 860,291 checks, plus a unit test with three vehicles following each other round a ring. Conditions: saturated demand, each road one resource; Okpo map before the floating docks and branch roads, Yantai map. Safe release mirrors signalmen granting passage in the field (KED Global, 26 Mar 2024); the banker's-algorithm safety check is borrowed from deadlock avoidance in operating systems (Habermann 1969, to be checked).

# Segmenting the dock road: main scenario, ceiling only (+8%); dock-bound erection mix, both ends (+76% / +8.3%)



## Exchange-rate side: the bound binds, road changes pay

Okpo erection mix, whole road: floor / ceiling = 0.87 (0.86 on the completed mix, 0.92 for the reference rule). Segmenting Okpo's dock-1 road lifts the ceiling by 76%: control granularity is a first-order item on the retrofit menu.

## Traffic-ordering side: the plateau is far below the bound, orchestrate first

Both main scenarios 0.34–0.36 (on the completed mix: reservation 36–45%, reference rule 50–57%); the same erection mix falls to 0.53 once segmented: change the granularity and the bottleneck moves from the road to the rule.

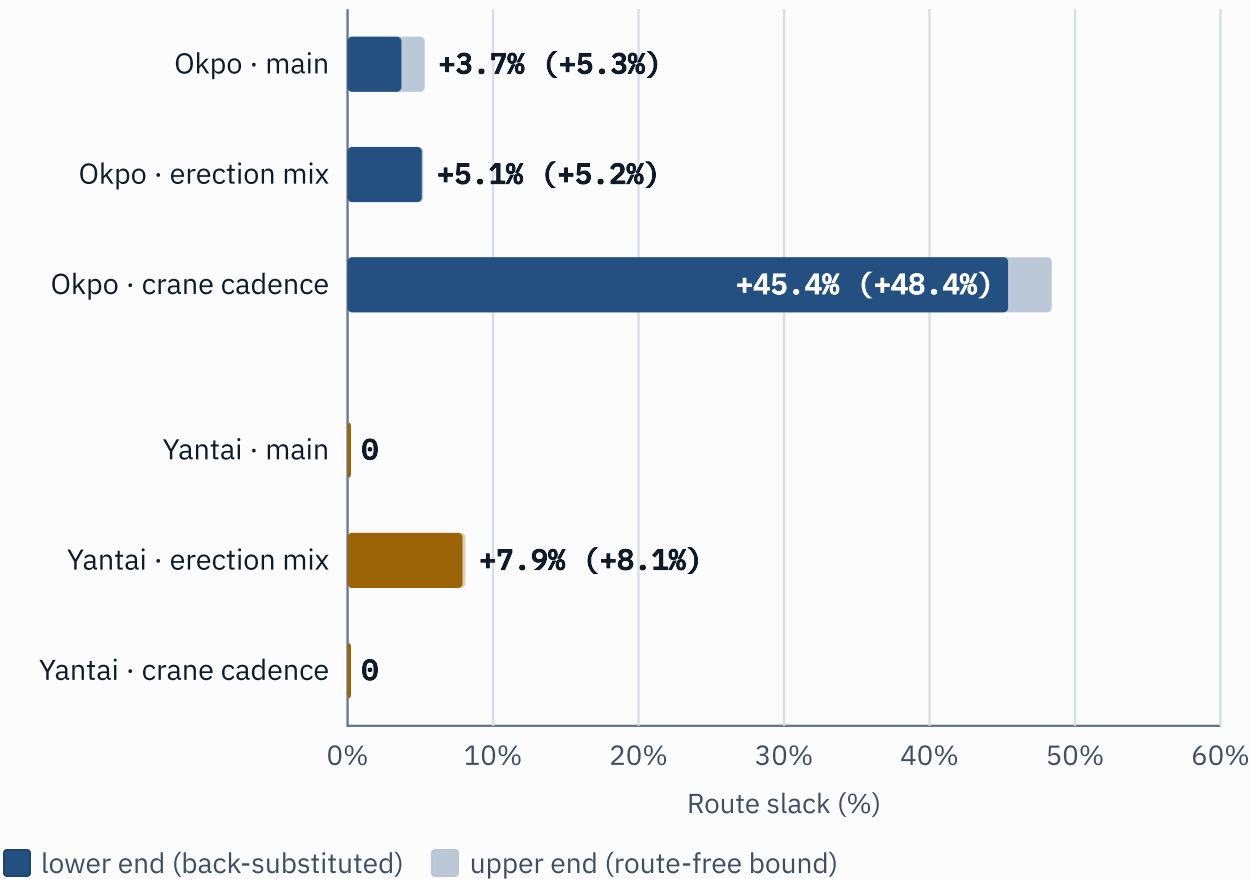
These are two ready examples of the boundary between the traffic-ordering and exchange-rate claims; retrofit simulations and the road–vehicle exchange-rate table will turn it into a decision rule.

Capacity interval Floor: reservation plateau safe-release plateau

Source: simulations in this study: both yards' main scenarios and Okpo's erection mix, whole and segmented roads (plateaus are means over  $K = 100$ – $150$ , recomputed and checked from each batch's curves by this deck's data script); the completed-mix caliber counted separately. Segmenting rule: roads longer than 200 m with roadside stops get block boundaries 25 m before and after each stop (one rule for both yards, a scenario setting).

Conditions: saturated demand; Okpo map before the floating docks and branch roads, Yantai map.

# A firmer ceiling: the noise-free system bound and the route-free bound; the route slack depends on what binds

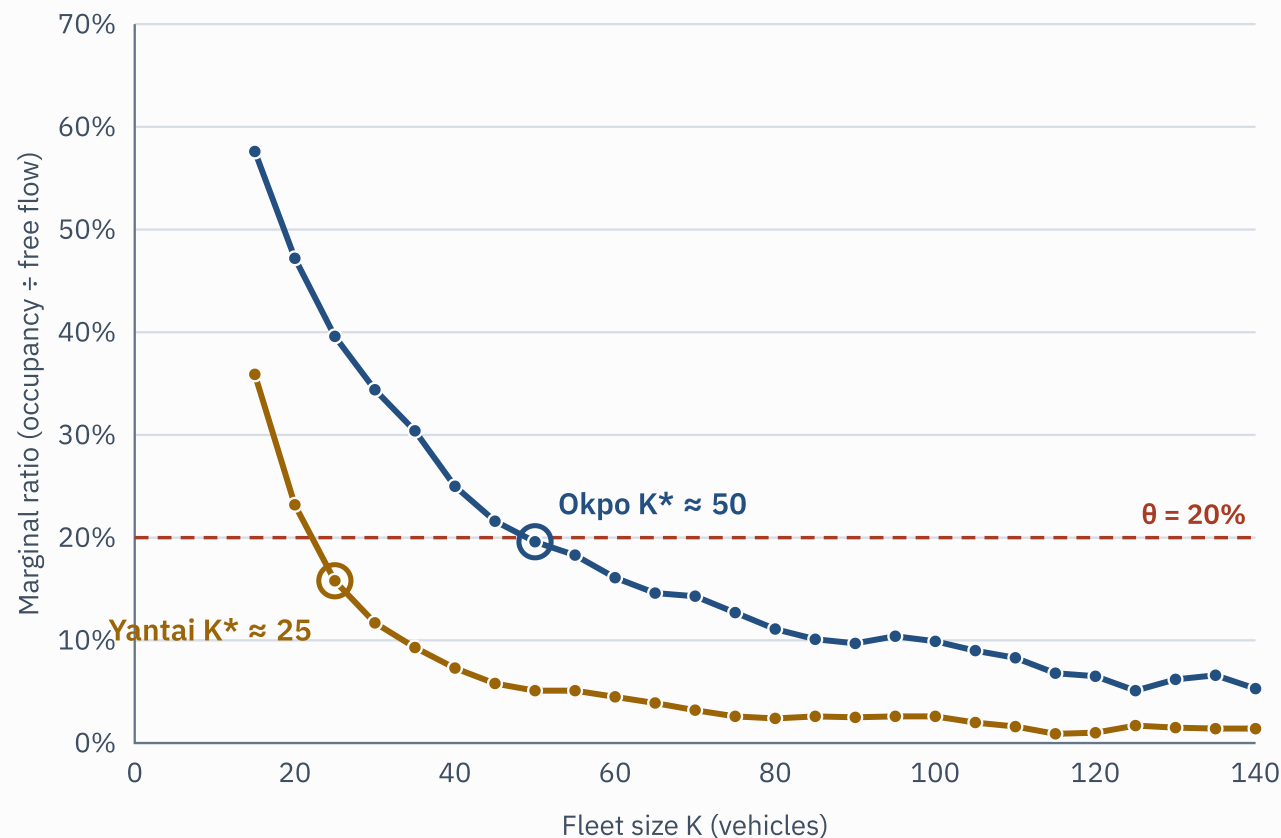


| Yard · scenario        | System bound | 10 seeds    | Route-free bound |
|------------------------|--------------|-------------|------------------|
| Okpo · main            | 1,736        | 1,436–1,948 | 1,828            |
| Okpo · erection mix    | 697          | 624–783     | 733              |
| Okpo · crane cadence   | 1,794        | 1,661–1,969 | 2,662            |
| Yantai · main          | 1,298        | 1,157–1,318 | 1,298            |
| Yantai · erection mix  | 1,110        | 971–1,212   | 1,200            |
| Yantai · crane cadence | 1,225        | 1,111–1,240 | 1,225            |

- **Entrances:** zero slack by construction (the binding entrance has a single incoming arc)
- **Dock stopping roads:** little through traffic (under 1% on the dock-1 road); the slack comes from entering at the other end, +3.7% to 7.9%
- **Through corridors:** under crane cadence 97% of the flow on one Okpo through corridor is passing through; routing round it gives +45.4%

Source: computations in this study: system bound, route-free bound and back-substituted feasible values for three scenarios on both yards. Route slack is an interval: lower end = back-substituted feasible value – system bound, upper end = route-free bound – system bound; no counterexample to the route-free bound has been found. The system bound constrains steady-state operation; chapter 3 quotes the system value and the case-by-case check keeps the per-seed network bound; a draft Jensen proposition: the harmonic mean of sampled bounds does not exceed the system value in expectation. The draft proofs of both bounds are being rewritten.

# Network-bound criterion $K^*(20\%)$ : from $K \approx 50$ on Okpo and $K \approx 25$ on Yantai, one more vehicle adds less than 20% of its free-flow gain



■ Okpo ■ Yantai —  $\theta = 20\%$

## $K \approx 50$

Okpo: network-bound from here (range 50–60)

## $K \approx 25$

Yantai: a small network starts much earlier

## 58% → 34%

Okpo,  $K = 15 \rightarrow 30$ : one more vehicle's gain as a share of free flow; usual fleet sizes are already in the transition

**Criterion:**  $K^*(20\%)$  is where the marginal throughput of one more vehicle first falls below 20% of free flow and stays below thereafter.

Source: simulations in this study: summaries of the tightened-bound round (Okpo) and the Yantai round (network identical to the current Yantai map); reproduced with the same definition in the physical-rules round (Okpo 50 / 100, Yantai 25 / 35). Marginal ratio = occupancy gain ÷ free-flow gain over a 20-vehicle window centred on  $K$ ; the range is read from the ratio's 95% interval; at  $\theta = 10\%$ : Okpo 100, Yantai 35. Conditions: whole-route reservation, saturated demand, each road one resource.

# The network-bound hypothesis in three parts: all have readings; the final wording of parts 2 and 3 follows the rule comparison

1

## The capacity interval and the start of the network-bound regime $K^*$

Okpo [627, 1,720],  $K^*$  50–60; Yantai [455, 1,252],  $K^*$  25 (whole-route reservation, saturated demand)

established

five simulation rounds; floor from reservation, re-checked with the new rule

2

## At whole-yard volume, where the vehicles needed sit relative to $K^*$ , and the gap to free flow

Okpo's own 339 a day: reservation needs 27 (16 h) / 15 (24 h), below  $K^*$ ; the peak of 678: 117 / 46; 600 a day: 88 / 37

readings in

new numbers and main caliber: after the rule comparison

3

## Whether the dock mouth becomes the yard's bottleneck depends on whether vehicles waiting for the crane block the way to other stops: where they wait, and whether the dock road is used in sections

Waiting off the road (reservation), throughput never drops; rules that wait on the road, teleport-free safe release included, drop with crane utilisation, the threshold moving left with fleet size and right with segmented blocking

readings in

two-layer wording: after the rule comparison

### Vehicles-needed criterion

Vehicles needed (mean throughput  $\geq 99\% \times$  daily volume)  $\geq K^*$  means network-bound

### Density criterion

Daily volume / network bound; volume at whole-yard caliber, converted the same way for both yards

### Conclusive either way

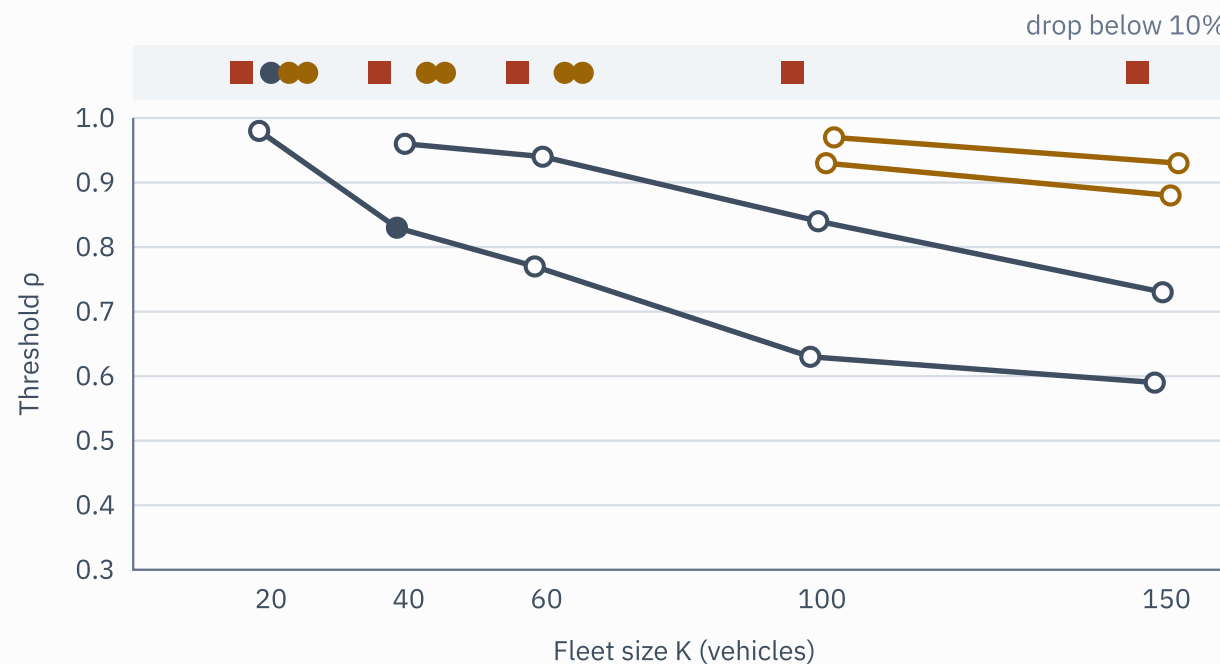
Holds: shipyard transport has network-bound capacity. Not: a density criterion bounds where free flow suffices

# Dock mouth (network-bound hypothesis, part 3): rules that wait on the road drop, earlier with more vehicles; reservation never drops

Okpo

Yantai

Crane-cadence  $p$  sweep, saturated, 8 days



■ Whole-route reservation (both) — Safe release · whole road - - Safe release · segmented

— Reference rule (teleport) · whole road - - Reference rule (teleport) · segmented

○ hollow = uses an unsteady point

**0.98 → 0.59**

Safe release, whole road: the threshold moves from 0.98 at  $K = 20$  to 0.59 at  $K = 150$  (Yantai 0.93 → 0.38); more vehicles, longer queues on the road

**0.83 → 0.96**

Okpo,  $K = 40$ , safe release: segmented blocking at the stops moves the threshold right

- **Crane starvation:** at  $p = 0.99$  safe release leaves the cranes 12–25 points less used; under reservation Okpo's cranes still reach 96%
- **Teleport contamination:** under the reference rule about 40–50% of truck-borne erection moves are teleported at least once on the way to the dock, so the dock conclusions rest on reservation and safe release

Source: simulations in this study: crane-cadence  $p$  against fleet size  $K$  on both yards (safe release on whole and segmented roads; whole-road reservation and reference rule reused from the gantry-span round). Threshold = the crane utilisation at which throughput first falls 10% below the lowest- $p$  level (Okpo 0.53, Yantai 0.13), interpolated between neighbouring levels; saturated, 8 days, 10 seeds; Okpo map before the floating docks and branch roads, Yantai map.

# Whole-yard caliber: Okpo's own 339 a day and peak 678; vehicles needed to the single vehicle, coloured by the network-bound test

- **Okpo itself:** converted as for Yantai (steel ÷ block weight × moves ÷ working days), 339 a day (294–363 across years); Hanwha Systems' 105,000 moves a year ÷ 300 days = 350, 3% apart; the peak is twice that, 678 / 700
  - **Large Korean yards:** Yim 2008, over 300 a day; Shen 2018, about 600; Hyundai Heavy, about 500 over 24 h. 97 / 194 is a four-transporter subset, kept for comparison
  - **Yantai itself:** 23 / 46 a day by the same method, 1–2 vehicles
- Test:** vehicles needed = the smallest integer K whose 10-seed mean throughput reaches 99% of the daily volume; at or above the top of the K\* range it holds, inside the range it is near the start.

Okpo · whole road   Okpo · segmented   Yantai   K\* = 50–60

| Tasks a day             | 16-h working day |       |       |       | 24-h working day |       |       |      |
|-------------------------|------------------|-------|-------|-------|------------------|-------|-------|------|
|                         | Free             | Resv. | Safe  | Ref.  | Free             | Resv. | Safe  | Ref. |
| 300                     | 15               | 22    | 34†   | 19    | 10               | 13    | 13    | 12   |
| 339 own mean            | 17               | 27    | 50†   | 22    | 12               | 15    | 15    | 14   |
| 350 direct anchor       | 18               | 29    | 66†   | 23    | 12               | 16    | 16    | 14   |
| 450                     | 23               | 47    | > 200 | 32    | 15               | 22    | 52†   | 19   |
| 500 Hyundai Heavy scale | 25               | 58    | > 200 | 37    | 17               | 27    | 68†   | 22   |
| 600 Shen et al. scale   | 30               | 88    | > 150 | 47    | 20               | 37    | 192†  | 27   |
| 678 own peak            | 34               | 117   | > 200 | 56    | 23               | 46    | > 200 | 32   |
| 700                     | 35               | 123   | > 150 | 59    | 24               | 49†   | > 150 | 33   |
| 900                     | 45               | > 150 | > 150 | 92    | 30               | 95    | > 200 | 47   |
| 1,200                   | 60               | > 150 | > 150 | > 150 | 40               | > 150 | > 150 | 73   |

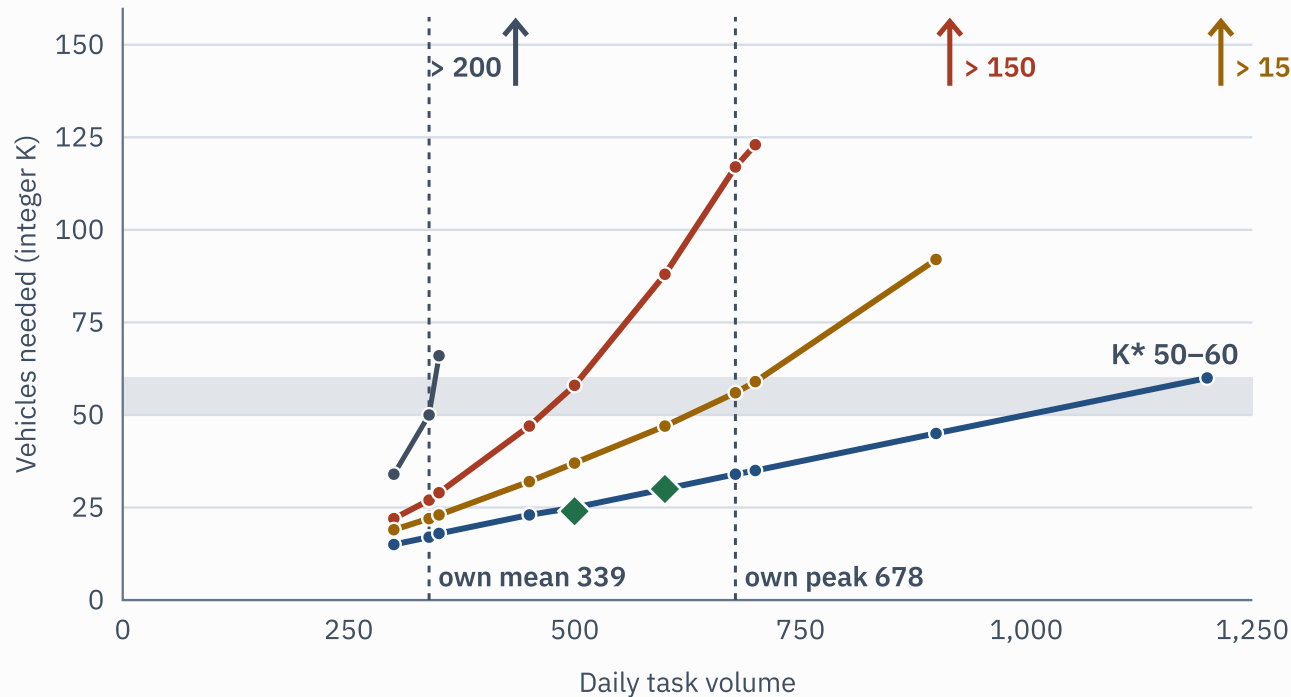
■ holds (≥ top of K\*)   ■ near the start (inside the K\* range)   □ does not hold (< bottom of K\*)

† a higher fleet falls short again (near the threshold); safe release “> 150” was not scanned further, “> 200” still short after scanning K = 100–200; Yantai's 23 / 46 have 16-h readings only

Source: simulations in this study: vehicles needed at whole-yard volumes, Okpo on whole and segmented roads and Yantai, 16-h and 24-h days; Yantai's 23 / 46 from an earlier round; Okpo's own volume converted in the external-anchor and literature check. Conditions: main-scenario flows, released in 8 batches a day, 4 days; Okpo map before the floating docks and branch roads, Yantai map.

# The work-zone chart: under a 24-h day, real fleets sit between free flow and whole-route reservation

Okpo Yantai 16 h 24 h



**17 < 24 < 27**

About 500 a day over 24 h: the model needs 17 in free flow and 27 under reservation; Hyundai Heavy's 24 sit between

**20 < 30 < 37**

About 600 a day over 24 h: free flow 20, reservation 37 (reference rule 27); Shen et al.'s about 30 sit between. Over 16 h it takes 30–88, and only free flow matches

Working-day length is a first-order parameter: Hyundai Heavy states 24 h, while Shen et al. and Okpo publish no shifts. Both days are shown; the main caliber is set after the rule comparison. The comparison takes a real fleet as the vehicles needed (no maintenance or spare margin), so it reads as magnitude and order.

■ Free flow ■ Whole-route reservation ■ Safe release ■ Reference rule (teleport) ■ K\*

◆ field fleets: Hyundai Heavy 24 (500 a day), Shen et al. about 30 (600 a day)

Sources: simulations in this study (whole road, integer K); field fleets: Park & Seo 2006 (Hyundai Heavy, 24 vehicles, 24 h, about 500 a day) and Shen et al. 2018 (about 600 a day, about 30 vehicles, shift not stated), both checked in full text. K\* is the main-scenario reservation range. With segmented dock roads, reservation's vehicles needed differ from whole roads by at most 1. Conditions: Okpo map before the floating docks and branch roads, Yantai map.

# The network-bound test under reservation: Okpo's mean is not network-bound, its peak and 600 a day are on 16-h days

| Volume caliber                                               | 16-h working day                 | 24-h working day                          |
|--------------------------------------------------------------|----------------------------------|-------------------------------------------|
| Okpo own mean (339 same method / 350 direct anchor) K* 50–60 | does not hold reservation 27–29  | does not hold reservation 15–16           |
| Okpo own peak (678 / 700, mean × 2) K* 50–60                 | holds reservation 117–123        | lower edge of the start reservation 46–50 |
| Hyundai Heavy scale (500 a day) K* 50–60                     | near the start reservation 57–58 | does not hold reservation 27              |
| Large Korean yard scale (600 a day) K* 50–60                 | holds reservation 88             | does not hold reservation 36–37           |
| Yantai own (23 / 46) K* 25                                   | does not hold reservation 1–2    | already below at 16 h                     |
| Yantai at Korean scale (450 a day) K* 25                     | holds reservation 69             | does not hold reservation 19              |

### The test

Vehicles needed at or above the top of the K\* range: holds; inside it: near the start; below it: does not hold. Each Okpo cell covers both granularities.

### Conditions

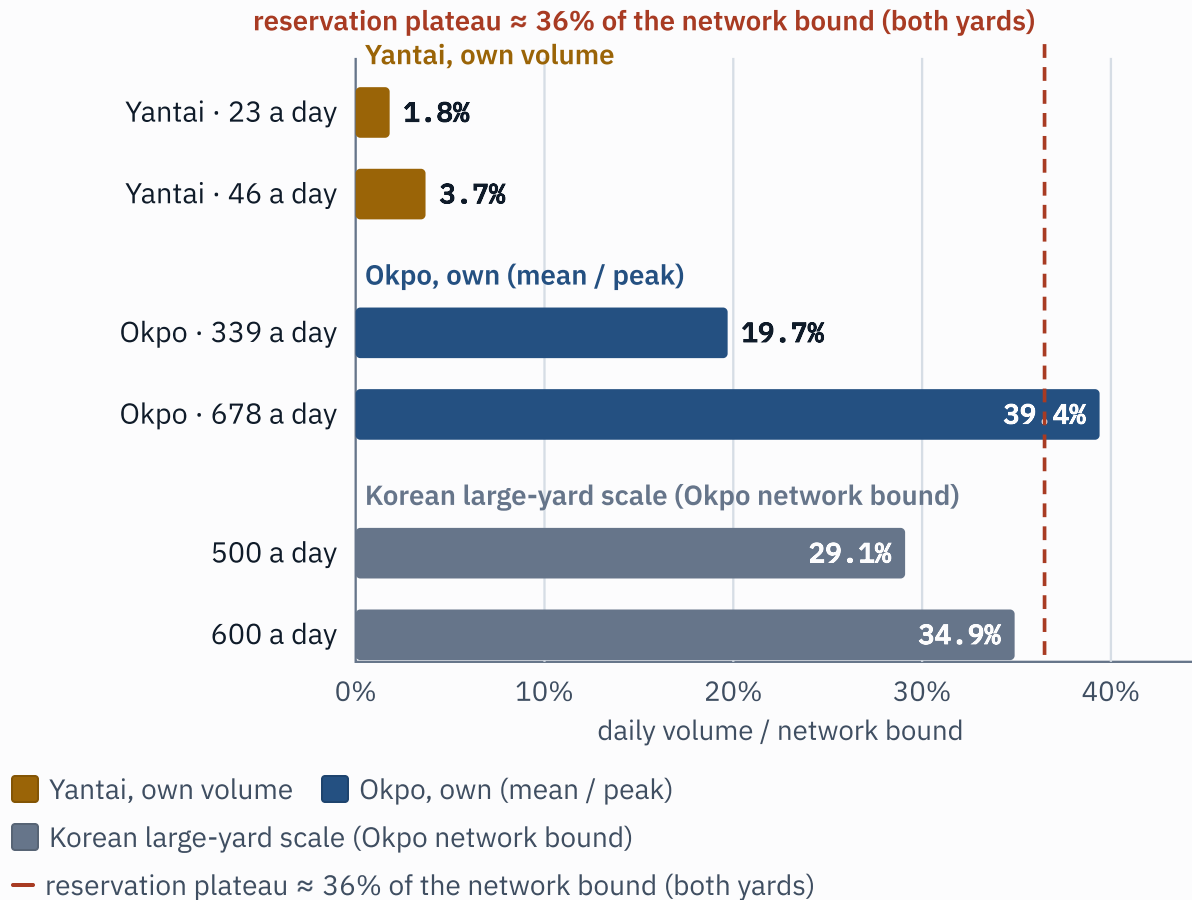
Whether the yard is network-bound depends on the volume caliber (mean or peak), the working day (16 h or 24 h) and the traffic rule; Okpo publishes no shifts, Hyundai Heavy states 24 h.

### Settled after the rule comparison

Part 2's new numbers, part 3's two-layer wording, and the main caliber (16 h or 24 h, mean or peak). This slide shows both calipers without choosing.

Sources: simulations in this study: vehicles needed and verdicts at whole-yard volume; K\* ranges from the tightened-bound round (Okpo 50 [50, 60]) and the Yantai round (25 [25, 25]); vehicles for Yantai's own 23 / 46 from an earlier round (16 h). Conditions: main-scenario flows; Okpo map before the floating docks and branch roads, Yantai map.

# Density puts the yards on two sides: Yantai's own volume is 2–4% of the network bound, Okpo's 20% (mean) and 39% (peak)



## Yantai: the side where free flow suffices

Its own 23–46 a day need 1–2 vehicles, far below  $K^* 25$ ; at a Korean large-yard scale of 450 a day (16 h), reservation would need 69, network-bound too.

## Okpo: below the regime at its mean, inside at its peak

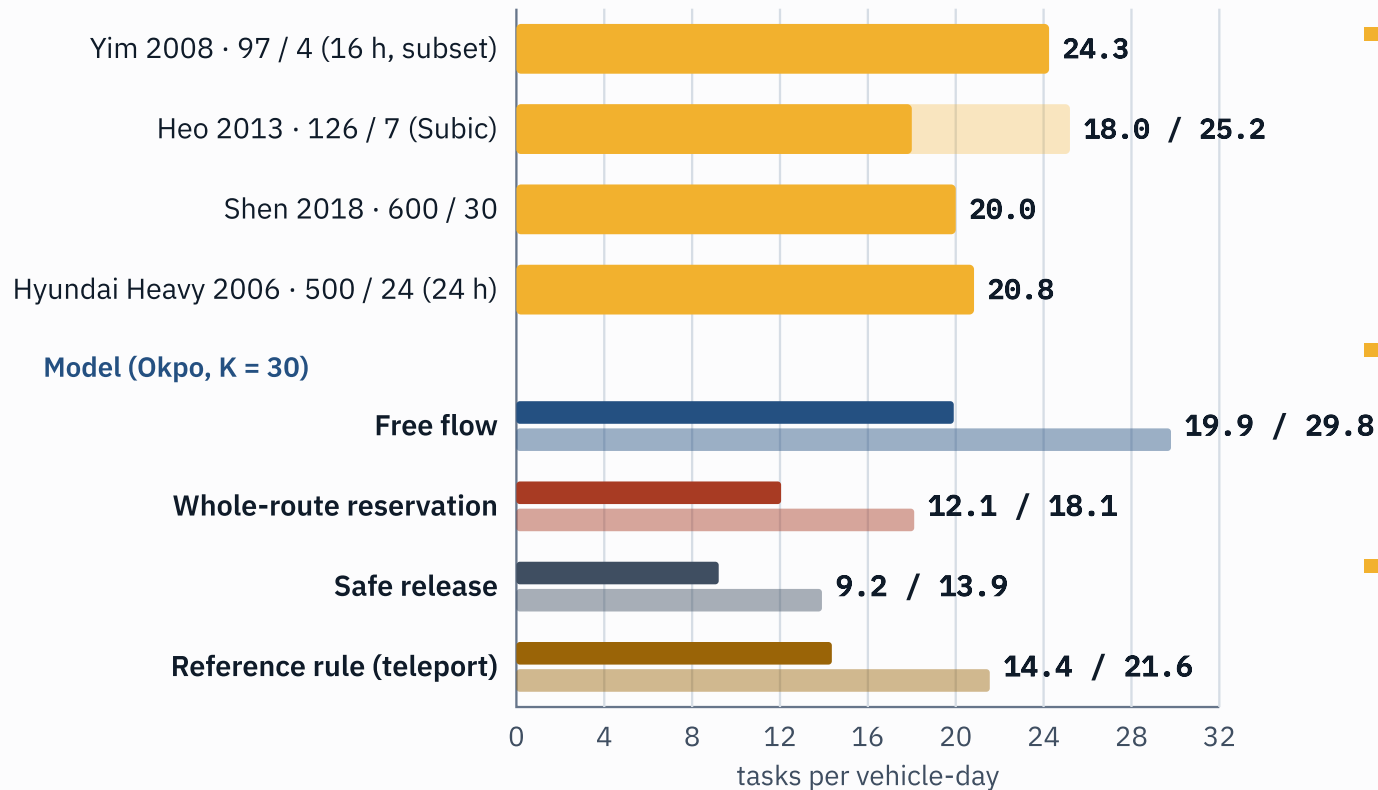
Its own mean of 339 is 20% of the network bound, with 27 vehicles under reservation on 16-h days; the peak of 678 is above the saturated reservation plateau of 627 and needs 117 on 16-h days (with finite demand idle vehicles stay off the road, the reservation table is emptier, and the volume is still reached).

Yantai's network bound binds at the two doors of what is presumably its paint shop (doors held exclusively while loading). The imagery shows vehicles can pass in front, so this granularity is conservative; the retrofit simulations add a door-granularity scenario.

Sources: computed for this deck as daily volume  $\div$  network bound (the network bound is the interval's ceiling, from this study's simulations); vehicles needed from this study's simulations (with finite demand idle vehicles stay off the road and the reservation table is emptier); Yantai's volume from an earlier round; binding element and imagery check as above (9 Oct 2026). Both yards' reservation plateaus are 36% of the network bound. The Korean large-yard scale is taken against Okpo's network bound, for reference only. Conditions: saturated plateau, each road one resource; Okpo map before the floating docks and branch roads, Yantai map.

# External anchors (full texts checked): 18–25 blocks per vehicle-day; normalise for cycle time and utilisation first

## Field anchors

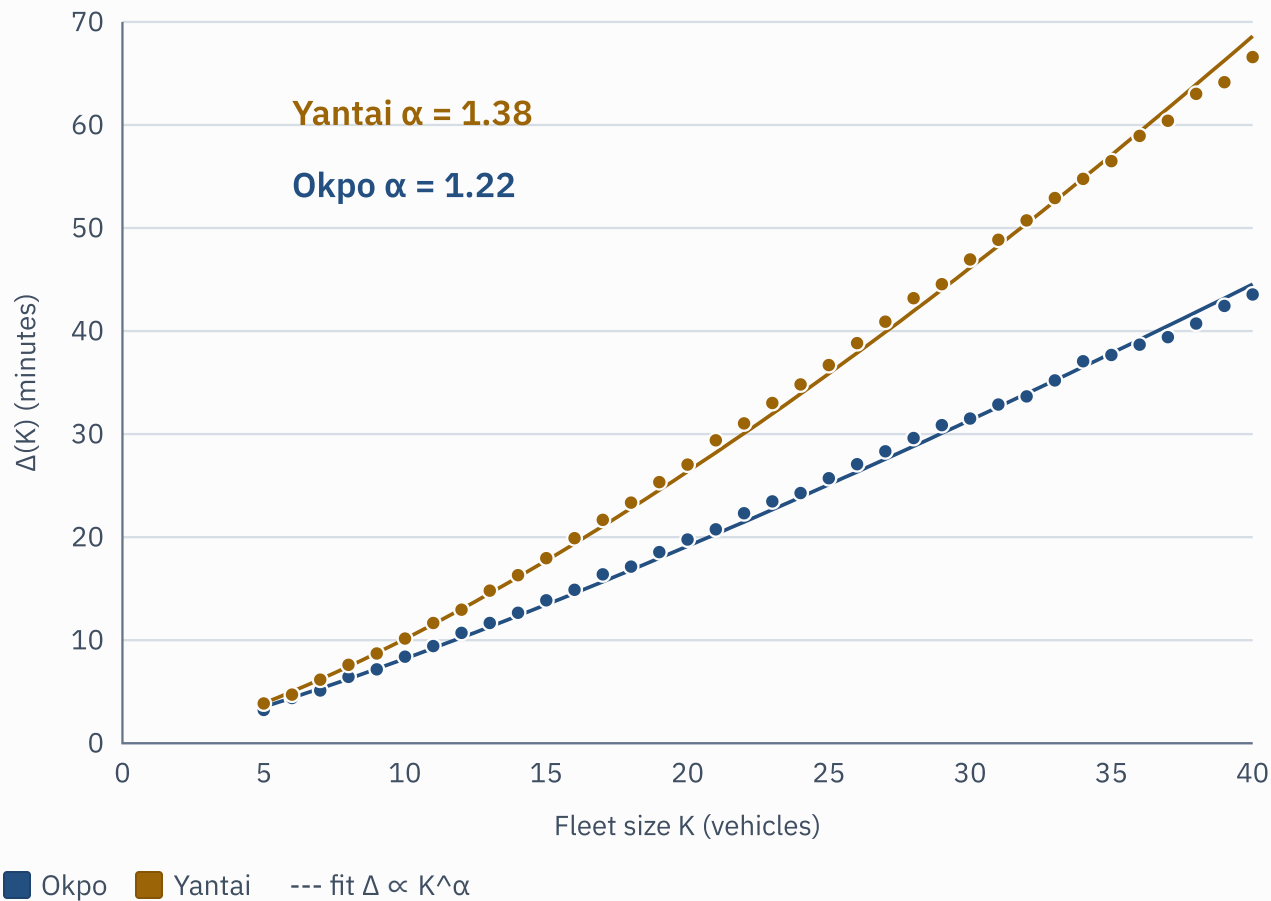


Field anchors 16 h 24 h

- Normalise:** output per vehicle = hours × busy share ÷ cycle. Okpo's model cycle without waiting is 48.3 min (empty 9.1, loading 10, loaded 19.1, unloading 10); Yim's 24.25 needs a cycle ≤ 39.6 min, yet those 4 vehicles were busy only about 27% of the day: the match at about 20 comes from cycle and utilisation cancelling
- Fleets:** Hyundai Heavy, 24 vehicles for about 500 a day over 24 h, against the model's 17 in free flow and 27 under reservation; Shen et al., about 30 for about 600 a day, against 20 and 37. Real fleets sit between the two (work-zone chart)
- Road load (next):** Song & Kang 2009 simulate 102–161 passes a day and up to about 26% use on roads around a stockyard; by Song and Kang's affiliation the yard is inferred to be DSME Okpo; to go into chapter 3's external comparison table against the model's per-road passes

Sources: Yim et al. 2008 (97 blocks / 4 vehicles, 8 a.m. to midnight), Heo et al. 2013 (Subic, 126 / 7; the light bar is 25.2 counting the 5 main vehicles), Shen et al. 2018, Park & Seo 2006 (Hyundai Heavy) and Song & Kang 2009, all checked in full text; model values from simulations in this study (Okpo, K = 30, saturated, 16 h and 24 h). Conditions: Okpo map before the floating docks and branch roads, each road one resource; the reference rule is for comparison only.

# Free-flow error: the service-time gap is superlinear ( $\alpha = 1.22 / 1.38$ ), and the vehicle ratio rises with volume



**Criterion:** service-time gap  $\Delta(K) \propto K^\alpha$  with  $\alpha > 1$ ; the ratio of vehicles needed is read at whole-yard volume.

**44 / 67 min**

Extra mean service time (dispatch to unloaded) of whole-route reservation over free flow at  $K = 40$ : Okpo / Yantai

| Tasks a day | 16 h: free → reservation | ratio | 24 h: free → reservation | ratio |
|-------------|--------------------------|-------|--------------------------|-------|
| 300         | 15 → 22                  | 1.5   | 10 → 13                  | 1.3   |
| 339         | 17 → 27                  | 1.6   | 12 → 15                  | 1.3   |
| 450         | 23 → 47                  | 2.0   | 15 → 22                  | 1.5   |
| 600         | 30 → 88                  | 2.9   | 20 → 37                  | 1.9   |
| 678         | 34 → 117                 | 3.4   | 23 → 46                  | 2.0   |
| 900         | 45 → > 150               | > 3.3 | 30 → 95                  | 3.2   |

Source: simulations in this study: mean service time of each batch at saturated demand,  $\Delta$  = reservation – free flow;  $\alpha$  is the log-log least-squares slope over  $K = 5\text{--}40$  (the same fit as in the experiment script). Okpo's 1.22 matches the experiment summary; fitted over all  $K$  (2–150), Yantai gives 1.29. Ratios of vehicles needed from the integer vehicles needed at whole-yard volume. Conditions: whole-route reservation (just in time), each road one resource; Okpo map before the floating docks and branch roads, Yantai map.

# Established: physical rules, system bound, external anchors; rule comparison running; retrofits, floating docks next

## Established (Okpo before floating docks and branch roads, Yantai; whole and segmented)

- **Bounds:** the network bound checked case by case on 57,917 runs, none above; the system bound removes sampling noise, the route-free bound holds for any routing rule
- **Capacity interval:** Okpo [627, 1,720] (segmented [630, 1,857]), Yantai [455, 1,252]; erection mix [611, 703] → [662, 1,241]
- **Best deadlock-free rule:** whole-route reservation; safe release is exact and reliable but conservative
- **Readings for all three parts of the network-bound hypothesis:**  $K^*$ ; integer vehicles needed (16 h / 24 h); the dock-mouth mechanism
- **Free-flow error:**  $\alpha = 1.22$  and  $1.38 > 1$ ; the vehicle ratio rises with volume
- **External anchors** (full texts): on 24-h days field fleets sit between free flow and reservation

## Still being settled (running or queued, not results)

- **Rule comparison (running):** a less conservative, still reliable deadlock-free rule; re-checks the floor, the decline and the safe-orchestration cost hypotheses
- **Open (after the rule comparison):** part 2's numbers and part 3's two-layer wording of the network-bound hypothesis, the main caliber
- **Retrofits (queued):** retrofit simulations and the exchange-rate table, door granularity, block-length curve, the three-resource dock network
- **Floating docks (queued):** floating docks; finite holding area  $S$  and  $K_0(S)$
- **Theorems:** draft proofs of the network and route-free bounds being rewritten; the safe-orchestration theorem tested in the rule comparison

# Next: the rule comparison, then three network-bound decisions, then retrofits and floating docks; each step with what it feeds

| STEP                                  | CONTENT                                                                                                                                                                                                                                                                                                                                                                                                                              | OUTPUT (PLACE IN THE THESIS)                                                                                                      | STATUS               |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|----------------------|
| Less conservative deadlock-free rules | Still reliable: the banker's check plus a step-wise feasibility search (grant if either holds), or claims cut at the next safe holding point (safe-orchestration theorem); report the share of waits from rejection; first switch to the common yard-map dataset (Okpo with floating docks and branch roads), close the direction-lock U-turn gap at stops, split reservation waits into crane and road, disambiguate resource names | Chapter 4 rule comparison; chapter 3 interval table, floor column; re-check of the decline and safe-orchestration cost hypotheses | running              |
| Three decisions                       | Decide together: part 2's new numbers and part 3's two-layer wording of the network-bound hypothesis, the main caliber (16 h or 24 h, mean or peak)                                                                                                                                                                                                                                                                                  | Final wording of the hypothesis; chapter 3 network-bound test table                                                               | after the comparison |
| Retrofits, exchange rate              | On the new rule's code and the Okpo map with floating docks and branch roads: retrofit simulations and the first road-vehicle exchange-rate table; capacity against block length; the three-resource dock network; Yantai's door granularity; road load in the external comparison                                                                                                                                                   | Chapter 5 exchange-rate table; the ordering / exchange-rate boundary rule; chapter 3 dock section and external comparison table   | queued               |
| Floating docks, holding areas         | Floating-dock scenario (quay-side sink and floating crane); finite holding area $S$ and $K_o(S)$                                                                                                                                                                                                                                                                                                                                     | The trapezoid's fall (decline hypothesis); chapter 5, what a holding area is worth in vehicles; the vehicle-type hypothesis case  | queued               |

Sources: the work plan of this study (decided 10 Oct 2026: the deadlock-free rule comparison before the retrofit simulations); the risks and strengthening plan from the physical-rules round.

# Risks and mitigations: each has a way through

| RISK                                                                                                                 | MITIGATION                                                                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Safe release claims conservatively, so the floor currently uses reservation                                          | The less conservative rules plug into the same batches and readings; the floor rises and the interval narrows, and the decline hypothesis is re-checked at the same time                                                                                             |
| Whether Okpo is network-bound moves with the caliber: not at its mean, yes at its peak and at 600 a day on 16-h days | The network-bound hypothesis is conclusive either way: holding at the peak shows network-bound capacity, not holding at the mean bounds where free flow suffices; both calipers are reported, the main one set after the rule comparison                             |
| Neither Okpo nor Shen et al.'s yard publishes shifts, and the fleet comparison depends on working-day length         | On 24-h days real fleets already sit between free flow and reservation (Hyundai Heavy states 24 h); normalise further by cycle time and utilisation (Hanwha Systems' 6 km mean haul as an input), as a column of the external comparison table                       |
| The 200 m and 25 m of segmented blocking are scenario settings, not field measurements                               | Whole and segmented roads are reported side by side; the retrofit simulations draw capacity against block length, turning one setting into a curve; passing places at the dock are checked on the imagery                                                            |
| Competition in the field: the closest domestic neighbour is Tao et al. 2017                                          | Tao 2017 read in full: a single-yard simulation counting road capacity in vehicles to compare vehicle types, with no blocking-time formalism, rule comparison, deadlock or bound; it does not narrow the claim and the introduction names it; submit soon as planned |

Sources: the risks and strengthening plan from the physical-rules round; the external-anchor and literature check (Tao 2017 read in full and separated point by point).

# Paper one (chapters 2–3: bound, capacity interval, dock mouth): target journals from the school's 2026 directory

## When transporters own the road: capacity bounds and operating regimes in shipyard block logistics

Working title; “and dock-mouth instability” may be added if the dock proposition holds

- **Model and bounds:** the network and route-free bounds, with width bin-packing and crane rows
- **Capacity interval and work zones:** integer vehicles needed, the  $K^*$  range and the density criterion, matched in magnitude to field fleets
- **Dock mouth:** where vehicles wait, how the road is sectioned, how many wait on it
- **The ordering / exchange-rate boundary and the first road–vehicle exchange-rate table** (retrofit simulations)

B-class first choice; the A-class journal is a stretch, with its condition on the right.

### First choice (B)

**Transportmetrica A: Transport Science (Class B) · Ocean Engineering (Class B)**

Transport science: occupancy model, bounds, rule comparison; naval engineering: two real yards, dock mouth, retrofits

### Stretch (A)

**Reliability Engineering & System Safety (Class A)**

Condition: the dock proposition proved as a stability / resilience result (blocking-induced starvation and recovery)

### Alternatives (B)

**IEEE Transactions on Automation Science and Engineering (Class B) · Applied Mathematical Modelling (Class B)**

T-ASE: capacity and release control in automated logistics; AMM: the LP bound and blocking model

### Supplementary (C)

**Computers & Industrial Engineering (Class C) · Journal of Ship Production and Design (Class C)**

Manufacturing-logistics and shipyard-production readers

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Data: simulations and computations in this study (the rounds from the flag experiment to the system bound; batch conditions in each slide's source line); the claims and hypotheses of this study; full-text notes from the external-anchor and literature check. Maps traced on public imagery; this briefing contains neither imagery nor network geometry.

# What this sub-study adds over the closest work in the field, and where to go next

1

## A bound that holds for any rule

The network bound makes occupancy a structural capacity constraint; the route-free bound holds for any routing rule. Roh & Cha 2011 treat interference as a soft term in the objective; in the search, no shipyard work has such a bound.

2

## Capacity as a measurable interval

Measured on two real yards and two granularities as [best deadlock-free rule, network bound]; the floor is 36% of the ceiling in the main scenario, and the erection mix shows the ordering / exchange-rate boundary.

3

## Regime criteria that work both ways

Integer vehicles needed, the  $K^*$  range and the density criterion sort yards into the two sides and match field fleets in magnitude: on 24-h days real fleets sit between free flow and reservation.

← Thesis briefing

Ceiling: network bound →

Floor: traffic orchestration →

Hypothesis part 3: dock mouth →

Fleet-sizing paper →

**Capacity is not one number but an interval: the traffic rule sets its floor, the network its ceiling.**

**Thank you.**

Sources: Roh & Cha 2011's "interference" as a soft objective term, checked in full text (9 Oct 2026); the closest work in the field, Tao et al. 2017, counts road capacity in vehicles to compare vehicle types, with no blocking-time formalism, rule comparison, deadlock or bound (full-text comparison with the closest work); search scope: shipyard transport scheduling, AGV zone control, lifelong multi-agent path finding. The eight task flows of the fleet-sizing paper, block sizes and vehicle parameters: see the flag-experiment setup.